10. Flatness–locality–geometry tradeoffs for Chern bands

Pure Thought research portfolio

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

In an ordinary energy band, particles can have quite different energies depending on how they move. A nearly flat band reduces those energy differences, potentially making interactions between particles more influential. We want that band to also have a nontrivial topological structure. The problem asks how well these goals can be combined when particles are allowed to hop only a limited distance and the model has a limited number of orbitals.

Key terms

- **Flat band** — A band whose energy is constant across momentum space; a nearly flat band has a small but nonzero variation.
- **Bandwidth W** — The difference between the largest and smallest energy in the selected band.
- **Gap Δ** — The minimum energy separation between that band and other bands.
- **Flatness ratio W/Δ** — A way to compare bandwidth to isolation from other bands; smaller values mean a flatter, well-separated band.
- **Hopping range** — The greatest lattice distance over which the model allows direct hopping.
- **Chern band** — A band with a nonzero Chern number, indicating nontrivial global topology.
- **Berry curvature** — A local measure of how quantum states twist as momentum changes; its integral determines the Chern number in the stated convention.
- **Quantum metric** — A measure of how distinguishable nearby momentum-dependent quantum states are.
- **Spectral flattening** — A mathematical change to band energies that preserves the chosen states but generally changes the spatial range of hopping.

Why this matters

Flat topological bands are useful settings for studying collective quantum behavior driven by interactions. But there are tradeoffs: restricting the model's spatial range changes which flatness properties are possible. Quantifying those tradeoffs can show whether a proposed improvement is attainable or requires a more complicated model.

What progress would mean

Progress would provide certified examples and limits within a defined design family. Favorable single-particle geometry would make a model interesting to investigate further; it would not by itself prove an interacting fractional topological phase.

Core question

For fixed orbital count, hopping range, and gap normalization, how small can a Chern band's bandwidth and geometric nonuniformity be, with certified bounds?

Scope and assumptions

Choose a two-dimensional isolated rank-one band, target C=1, and a compact coefficient family with range R and fixed orbital positions. Initially use a specified two-band family extending Challenge 09. Impose a positive gap floor and a norm bound to prevent trivial energy rescaling.

Maintain two distinct tracks: finite-range approximate flatness, and exact spectral flattening with generally non-finite-range hopping. Do not require exact nonzero-Chern flatness with strictly finite-range hopping under the no-go theorem's hypotheses.

Mathematical target

Define W=max_k E_n−min_k E_n and Δ=min_k min_{j≠n}|E_j−E_n|. Optimize W/Δ. For the fixed Brillouin-zone coordinates define g_ij=Re$\langle\partial$_iu|(1−P)|∂_ju$\rangle$ and F with the same Chern convention as Challenge 09. Then C=$(1/2\pi)\int$F and $\bar{F}=2\pi$C/Area(BZ).

Report separately Var(F)=Area$^{-1}\int$(F−$\bar{F}$)^2 and D_g=Area$^{-1}\int$(tr g−|F|). These are diagnostics under a specified Euclidean coordinate metric, not a universal fractional-Chern-insulator stability score.

Required outputs and proof obligations

Export coefficient/domain constraints, W upper and Δ lower bounds, certified C, geometric enclosures, and optimization gaps. An achieved objective gives an upper bound on the minimum; a global lower bound needs a covering/relaxation proof. For infinite-range flattening, give decay and truncation estimates.

Validation and rejection controls

Verify normalization $\int$F=2πC and the pointwise metric inequality in the chosen convention. Check that truncating a spectrally flattened model generally reintroduces bandwidth. Include a trivial exactly flat band as a negative topology test. Compare candidate objectives using identical units, embedding, and norm constraints.

Milestones and research extension

Baseline: one certified near-flat Chern model and a correct locality audit. Strong: global objective bounds on one compact family or a rigorous truncation tradeoff. Research extension: optimal geometry within a specified class; many-body FCI stability requires an added interacting Hamiltonian and filling.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Chen et al., Flat Chern-band locality obstruction — No-go theorem for simultaneous exact flatness, nonzero Chern number, and strictly local hopping.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$W = \max_k E_n(k) - \min_k E_n(k), \quad \Delta = \min_k \min_{j\neq n} |E_j(k) - E_n(k)|, \quad \text{minimize } W/\Delta.$$

$$g_{ij} = \mathrm{Re}\langle \partial_i u | (1-P) | \partial_j u \rangle, \quad \bar{F} = \frac{2\pi C}{A}, \quad A = \mathrm{Area(BZ)},$$

$$\mathrm{Var}(F) = \frac{1}{A}\int_{\mathrm{BZ}} (F-\bar{F})^2 d^2k, \qquad D_g = \frac{1}{A}\int_{\mathrm{BZ}} (\mathrm{tr}\, g - |F|) d^2k.$$

Fix a gap floor, norm bound, orbital positions and coordinate metric. Use the Chern convention of 09. Strictly finite-range approximate flatness and generally long-range exact flattening are separate tasks; these diagnostics alone do not prove an interacting fractional phase.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — incompatible design requirements (33–40; synthesis finite-range search).** Exact flatness, nonzero Chern number, and strictly local finite-range hopping cannot all be required under the cited theorem's assumptions. The rewrite splits the problem into two valid tracks.
2. **Blocking — normalization (38, 86).** The two displayed constant-curvature formulas disagree and omit/misplace 2π. Defining C by its integral fixes $\bar{F}$ uniquely.

3. **Major — unsupported objective (22–24, 39–40).** The stated stability ratio is not a general criterion proving an FCI ground state. Geometry, bandwidth, and an interacting many-body gap are different claims.
4. **Major — exact versus approximate (787–805).** A tiny numerical bandwidth or curvature variance does not establish exact flatness or uniformity. The revision asks for certified enclosures and distinguishes achievable values from global optima.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For fixed orbital count, hopping range, and gap normalization, how small can a Chern band's bandwidth and geometric nonuniformity be, with certified bounds?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: one certified near-flat Chern model and a correct locality audit. Strong: global objective bounds on one compact family or a rigorous truncation tradeoff. Research extension: optimal geometry within a specified class; many-body FCI stability requires an added interacting Hamiltonian and filling.

Sources supporting the corrections

- Chen et al., Flat Chern-band locality obstruction — No-go theorem for simultaneous exact flatness, nonzero Chern number, and strictly local hopping.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented

trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means “not excluded”; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.

- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.